# Bridging the Sim-to-Real Gap: Reinforcement Learning for Recommender Systems via *In Silico* LLM Personas

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October 20, 2026

# Grounding: Recommendation as Sequential Control

## What Recommenders Do Today: Myopic Ranking

- Platforms (Netflix, TikTok, YouTube) optimize for **immediate clicks or watch time**.
- **The Core Pitfall:** Greedy ranking traps users in repetitive filter bubbles, triggers rapid genre fatigue, and accelerates session abandonment.

## What We Aim to Solve: Long-Horizon Sequential Policies

- We formulate recommendation as a **sequential decision process**: balancing immediate engagement with long-term retention (managing satiation, encouraging serendipity).
- **The RL Dilemma:** RL is built for this, but live A/B testing is too risky, offline logs have exposure bias, and toy simulators have a severe **sim-to-real gap**.

# Existing Paradigms & The Simulation Gap

## Simulators & Offline RL

- **Parametric Simulators (RecSim):** Hand-crafted linear decay rules; policies overfit and fail on real users (*sim-to-real gap*).
- **Offline RL (e.g., CQL):** Penalizes unseen actions in static logs; suffers high variance and pessimism on sparse data.

## Generative User Agents

- **LLM Agents (Agent4Rec):** Encode rich cultural priors and simulate believable human ratings from history.
- **The Missing Link:** Treated strictly as **static evaluation benchmarks**, never interactive RL environments due to latency ($\sim 450$ ms/step).

# Proposal: *In Silico* Personas in a POMDP

## Recommendation POMDP (Grounded on MovieLens)

- **Empirical Grounding:** Environment grounded on MovieLens histories with rich metadata (titles, genres, ratings).
- **Hidden State ($s_t$):** Long-term user taste profile + transient session fatigue (unobservable).
- **Observed History ($o_t$):** Latent user profile embedding + recent interaction history.

## Proposed Interactive Training Loop:

1. **Continuous Action:** Policy outputs latent vector $\boldsymbol{a}_t \rightarrow$ mapped to items via cosine similarity.
2. **LLM Persona Step:** Processes context & history $\rightarrow$ returns rating $r_t \in \{1, \ldots, 5\}$ and retention $c_t \in \{0, 1\}$.
3. **Step Reward:** $R_t =$ Normalized Rating $+ 0.25 \cdot \mathbb{I}(c_t = 1)$ (satisfaction + retention).

# Method: Quantifying & Validating User Personas

## Proposed Behavioral Signatures

Condition personas on real user history across 3 conceptual dimensions:

- **Genre Preferences:** Empirical preference distribution over film genres.
- **Satiation & Fatigue:** Modeled sensitivity to repeated recommendations.
- **Popularity Tendency:** Preference for mainstream hits vs. niche indie cinema.

## Pre-RL Fidelity Validation (Planned on Held-Out Logs):

- **Rating Calibration:** Verify simulated ratings match real test rating distributions.
- **Distributional Alignment:** Quantify preference overlap with real user choices.
- **Temporal Stability:** Measure consistency across rollouts to ensure personas do not drift.

# Fast Token-Logit Scoring & KV Caching

## Proposed Zero-Generation Token Scoring

- Format prompt so the very next token is strictly the rating $\in \{\text{"1"}, \ldots, \text{"5"}\}$.
- In a **single forward pass** ($\texttt{max_tokens} = 1$), compute exact probabilities:

$$P_{\text{LLM}}(r_t = k \mid \text{context}) = \frac{\exp(\ell_k/\tau)}{\sum_{j=1}^{5} \exp(\ell_j/\tau)}, \quad k \in \{1, \ldots, 5\}$$

## Proposed Prefix Caching Across Parallel User Archetypes

- We plan to cluster users into representative archetypes pinned to parallel environments.
- System prompt KV cache hit rate $= 100\%$ via $\texttt{vLLM}$ prefix caching.
- **Expected Throughput:** Slashes latency from $\sim 450$ **ms** $\rightarrow <$ $12$ **ms** ($> 35\times$ speedup), making $10^6$ online RL steps computationally feasible on Yale's Bouchet HPC.

# Evaluation Protocol & Planned Experiments

## Planned Three-Way Policy Benchmark

We will train and compare three distinct paradigms:

- **Parametric Baseline:** Continuous-action SAC trained in **RecSim** (hand-crafted rules).
- **Offline Baseline: Conservative Q-Learning (CQL)** trained on historical logs.
- **Proposed Generative Setup:** Continuous-action SAC trained in $\mathcal{M}_{\mathbf{LLM}}$ with in silico personas.

## Planned Off-Policy Real Evaluation:

- **Sim-to-Real Transfer:** Evaluate all policies on held-out user trajectories to test whether LLM personas reduce downstream policy degradation.
- **Dense Matrix Fallback:** Use fully observed interaction matrices (KuaiRec) if MovieLens test splits exhibit high evaluation variance.

# Summary: Key Takeaways & Planned Roadmap

## Project Summary: What We Propose

- **Core Problem:** Classical RecSys simulators rely on toy rules that fail to capture human behavior, creating a severe sim-to-real gap.
- **The Approach:** Recast recommendation as a POMDP where empirical-grounded LLM personas act as dynamic human simulators.
- **Key Enabler:** Single-pass token-logit scoring + prefix caching ($< 12\,\text{ms}$) makes continuous online RL training computationally practical.

## Next Steps & Execution Plan:

1. Deploy `vLLM` on Yale's Bouchet HPC and validate persona fidelity on MovieLens.
2. Train baseline agents in RecSim and offline CQL.
3. Train SAC inside $\mathcal{M}_{\text{LLM}}$ and evaluate sim-to-real policy transfer.